

BAYES' THEOREM & RANDOM VARIABLES

Bayes' Theorem

Based on **Conditional Probability**:

$$P(A|B) = P(A \cap B) / P(B) \rightarrow A \text{ and } B \text{ happens} / P(B)$$

Probability of A given "B" is the area that is part of A in B.

Meaning: out of everything in B, how much is also in A.

$$\begin{aligned} P(B|A) &= P(A \cap B) / P(A) \\ \rightarrow P(A \cap B) &= P(A|B) \cdot P(B) \\ \rightarrow P(A \cap B) &= P(B|A) \cdot P(A) \end{aligned}$$

Bayes' Formula:

$$P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

Random Variables (RV)

Mapping random process outcomes into numbers. Denoted by **Upper Cases**.

Example: Fair coin. If Head (+), If Tails (-).

Types of Random Variables:

1. Discrete	2. Continuous
- Finite possible values	- Infinite possible values
- Prob value between 0 and 1	- Probability of each distinct value is 0
- Sum of all probabilities = 1	- Area under the curve = 1 (Integration)

Continuous Example: Probability of rain for *exactly* 3 minutes $P(X=3)$ is impossible (0) because it could be slightly more or less. Instead, we use intervals: $P(29 < X < 31)$.

PDF: Probability Density Function

- **Discrete:** Probability and Variance are the **Summation (Σ)**.
- **Continuous:** Probability and Variance are the **Area under the curve**.

Mean & Mode

- **Mean:** In Discrete, it is the *weighted mean*. In Continuous, there isn't a specific formula as it differs between distributions.
- **Mode:**
 - In Discrete: The value with the highest frequency.
 - In Continuous: The point where the probability is most likely to happen (Maximum value of the function).

How to find Mode in Continuous RV?

1. Find the first derivative: $f'(x) = 0$ to find the critical point.
2. Find the second derivative: $f''(x)$.
3. If $f''(x) < 0$, then it is a maximum (Mode).

Binomial Random Variable

Count of number of successes in a binomial experiment (e.g., rolling dice).

Requirements:

- Fixed sample size (n).
- Each trial can either succeed or fail.
- Probability for each event is the same.
- Each trial must be independent.